1. **9. Translate a problem into a standard mathematical representation or algorithm**
   - Translating the year-over-year population dynamics into a recurrence formula $N_{t+1} = 1.5N_t \times 0.8$, then into an exponential growth model $P = N_0 \times (1.2)^t$, and matching it to option B.
2. **5. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
   - Recognizing the fixed proportional growth pattern each year (growth factor 1.2), identifying it as a geometric sequence/exponential model.
3. **1. Select a mathematical description or a representation that describes a problem**
   - Choosing to use recurrence relations and exponential formulas to describe multi-year changes rather than enumerating year by year.
4. **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
   - Understanding the implication of "one-year-olds also breed" on the model, i.e., all individuals present at the start of the year participate in breeding, so the breeding increase is based on the total number at the start of the year.
5. **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
   - Corresponds to exponential growth/decay models in population dynamics, geometric sequence general term formulas.
6. **12. Identify the constraints, assumptions simplifications in a mathematical model**
   - The model relies on several assumptions (pairs form at the start of each year, each pair raises one chick, fixed mortality rate, one-year-olds breed, no other deaths or migration). The problem requires deriving the formula based on these assumptions.
7. **10/11. Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem**
   - Using N_t, recurrence formulas, and exponential functions to represent the structure.

1. **9. Translate a problem into a standard mathematical representation or algorithm**
   - 将种群动态年复一年的变化转化为递推公式 $N_{t+1} = 1.5N_t \times 0.8$，再转化为指数增长模型 $P = N_0 \times (1.2)^t$，并匹配到选项 B。
2. **5. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
   - 识别出每年固定的比例增长模式（增长因子 1.2），发现是等比数列/指数模型。
3. **1. Select a mathematical description or a representation that describes a problem**
   - 选择用递推关系与指数公式来描述多年度变化，而非逐年列举。
4. **4. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
   - 需理解“1 岁也繁殖”对模型的影响，即所有年初的个体都参与繁殖，因此繁殖增加量基于年初总数。
5. **8. Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
   - 对应人口指数增长/衰减模型、等比数列通项公式。
6. **12. Identify the constraints, assumptions simplifications in a mathematical model**
   - 模型基于多个假设（年初配对、每对养 1 雏、固定死亡率、1 岁繁殖、没有其他死亡或迁移），题目本身就是要求根据这些假设推导公式。
7. **10/11. Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem**
   - 使用 N_t、递推公式、指数函数来表示结构。